Electromagnet & VNA Control -- User Manual

Graphical user interface (GUI) guide for the electromagnet + Vector Network Analyzer measurement station (electromagnet equipment **YP180829-188-2**).

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> All screenshots in this manual were taken in **Simulation Mode** with made-up demo data (a synthetic calibration and a simulated sample), so you can follow every step without hardware. Your real numbers will differ.

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1. What the application does

The application drives the electromagnet's power supply and the VNA over GPIB and automates the whole measurement:

1. It converts the **magnetic field** you ask for (in oersted, Oe) into the **coil current** (in amperes, A) needed to produce it, using a current-to-field **calibration** you load or create.
2. For every field point it ramps the current there safely, waits for it to settle, triggers a VNA sweep, and reads back **S11, S21, S12 and S22** (magnitude *and* phase).
3. It plots the results live, saves every point to disk the moment it is measured, and keeps color maps (magnitude vs. field and frequency) up to date automatically.

A **maximum-current limit** that you set first is enforced on every single current command in the program. There is no override (see Tab 1).

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2. Starting the application

From the project folder, with the project's virtual environment active:

```
python -m src.main
```

(If you were given a packaged `ElectromagnetVNAControl.exe`, just double-click it. See `docs/PACKAGING.md` for how it is built.)

The application starts in **Simulation Mode**: instruments are simulated in software, so nothing real is driven. This is the safe way to learn the program or to check a sweep before running it on the magnet. To use the real hardware, see Tab 2.

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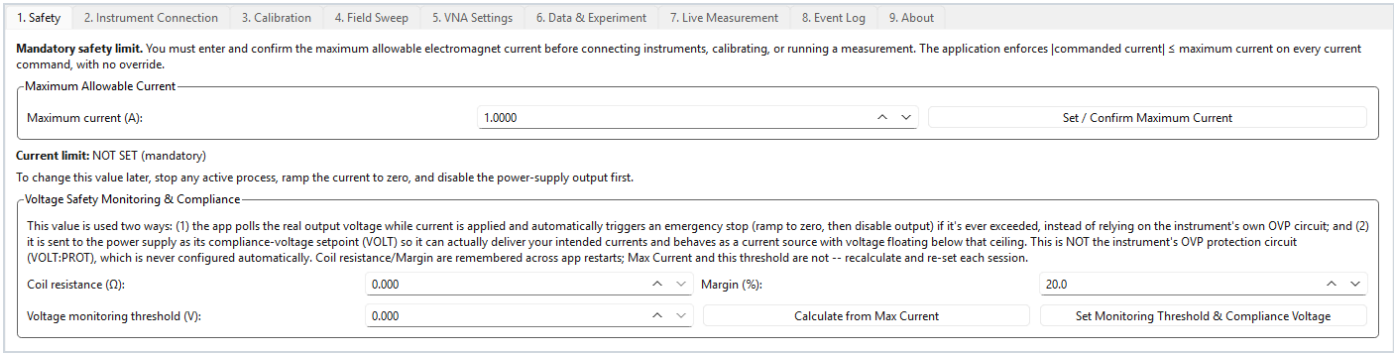
3. Window layout and the tab order

The window has nine tabs across the top. They are numbered in the order you normally use them, left to right:

#	Tab	Purpose
1	Safety	Set the mandatory maximum current; set voltage monitoring/compliance
2	Instrument Connection	Connect to the power supply and VNA (or simulation)
3	Calibration	Current -> field table (load, or build one point at a time)
4	Field Sweep	Define the fields to measure; check they are all reachable and safe
5	VNA Settings	Frequency range, power, bandwidth, which S-parameters
6	Data & Experiment	Name the experiment, create its folder, see the color maps

#	Tab	Purpose
7	Live Measurement	Run the sweep; live plots; background measurement; stop buttons
8	Event Log	Running log of everything the program does
9	About	Credits

Tabs 2-7 are greyed out until you set the maximum current on tab 1. Safety, Event Log and About are always available.



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4. Quick start: a complete experiment in 10 steps

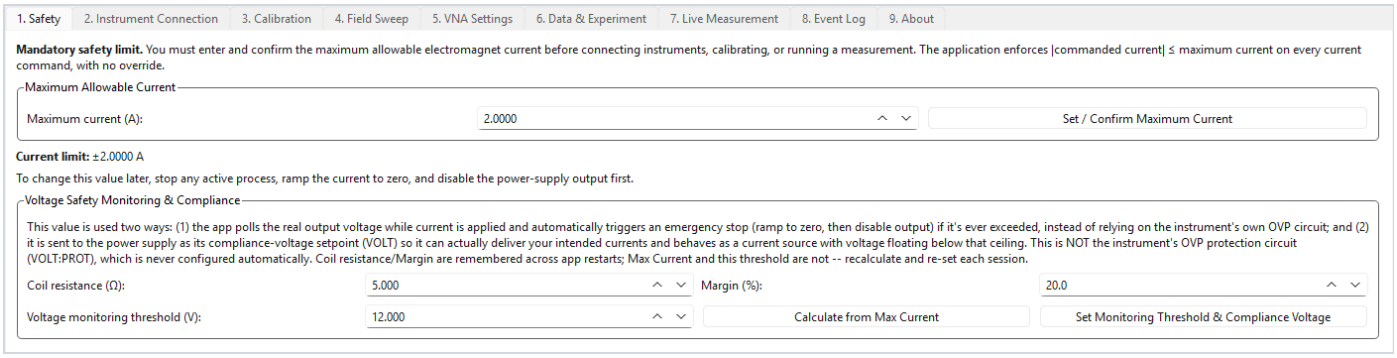
Use Simulation Mode first. Each step names the tab.

1. **Safety** -- enter the maximum allowable current (e.g. `2.0` A) and click **Set / Confirm Maximum Current**. The other tabs unlock.
2. **Instrument Connection** -- leave **Simulation Mode** checked (or uncheck it for real hardware) and click **Connect** for the Power Supply and for the VNA.
3. **Safety** (again) -- enter the **coil resistance**, click **Calculate from Max Current**, then **Set Monitoring Threshold & Compliance Voltage**.
4. **Calibration** -- load a calibration file (**Browse...** then **Load Calibration**) or build one (details).
5. **Field Sweep** -- choose mode, start, stop and step in oersted, then click **Build & Validate Sequence**. Every row must say `OK` and the summary must say **ALL VALID**.
6. **VNA Settings** -- set the frequency range and number of points, then click **Apply Configuration to VNA** (*after* the VNA is connected).
7. **Data & Experiment** -- type an experiment name, sample name and operator, then click **Create Experiment Folder**.
8. **Live Measurement** -- optionally click **Measure Background (0A)** to save a zero-field reference.
9. **Live Measurement** -- click **Start Measurement** and confirm. Watch the four S-parameter plots fill in.
10. When the "Measurement Finished" box appears, your data is already in the experiment folder (path shown on the Data tab and in the Live status panel). The output is ramped to 0 A and disabled automatically.

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5. Tab reference

Tab 1 -- Safety



Purpose: set the one number that protects the magnet and supply, and configure voltage monitoring.

Maximum Allowable Current

Control	What it does
Maximum current (A)	The absolute ceiling. Every commanded current, in every tab, must not exceed this value in magnitude (positive or negative).
Set / Confirm Maximum Current	Stores the limit and unlocks tabs 2-7. If the power supply is already connected it also writes the same limit to the instrument's own current limit (best-effort).

The status line shows `Current limit: ±2.0000 A`, or `NOT SET (mandatory)` until you confirm one. A current command above the limit is *rejected*, not clipped -- nothing is sent to the supply.

You cannot change the limit while a process is active or the output is enabled. The fields grey out and the status line says why (e.g. `Locked ... measurement_running`). To change it: stop the calibration or measurement, click **Ramp Current to Zero**, and make sure the output is disabled; then the limit can be edited again.

> The maximum current is **not remembered** between sessions on purpose -- you re-enter it every time you start the program.

Voltage Safety Monitoring & Compliance

This box does two jobs with one number:

1. **Software monitoring ceiling.** While current is applied, the app reads the supply's real output voltage and triggers an *emergency stop* if it ever exceeds this value.
2. **The supply's compliance-voltage setpoint (`VOLT`).** A constant- voltage/constant-current supply can only deliver the current you command if its voltage setpoint is high enough to regulate on current. The same value is therefore sent to the instrument, so it behaves as a current source with the actual voltage floating below this ceiling.

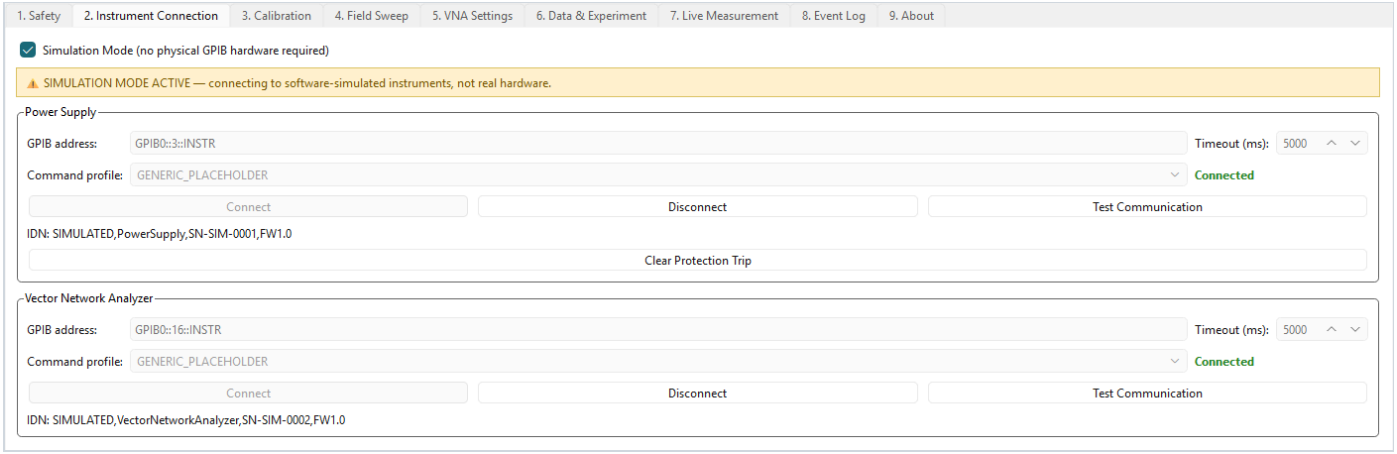
It is **not** the instrument's own over-voltage protection (OVP); the app never configures that.

Control	What it does
Coil resistance (Ω)	The electromagnet coil's DC resistance. Remembered between sessions.
Margin (%)	Head-room added on top of `I x R`. Remembered between sessions.
Voltage monitoring threshold (V)	The value that will be used. Filled by the button below, or type your own.
Calculate from Max Current	Computes `max current x coil resistance x (1 + margin/100)`. Example: 2 A x 5 Ω x 1.20 = **12 V**. Nothing is sent yet.
Set Monitoring Threshold & Compliance Voltage	Stores the threshold and, if the power supply is connected, sends it to the instrument as `VOLT`. A dialog confirms what happened.

Notes

- A threshold of 0 V is refused: it would stop any current from flowing. This almost always means the coil resistance is still 0.
- Connecting a new GPIB session resets the supply's voltage setpoint. The app therefore **re-applies the threshold automatically every time the power supply connects**, as long as you have set a non-zero one.
- The threshold is not remembered across restarts. Recalculate and re-set it each session.

Tab 2 -- Instrument Connection



Purpose: connect the power supply and the VNA.

Control	What it does
Simulation Mode checkbox	Checked: connecting creates software-simulated instruments. A yellow banner is shown while it is on. Unchecked: connecting talks to real GPIB hardware. **Disconnect both instruments before changing this.**
GPIB address	The VISA resource string, e.g. `GPIB0::3::INSTR` (power supply) and `GPIB0::16::INSTR` (VNA). See `docs/GPIB_ADDRESS_GUIDE.md` to find yours.
Timeout (ms)	How long to wait for an instrument to reply (100-60000 ms).
Command profile	Which set of SCPI commands to use.
Status	Disconnected (grey), Connecting (amber), Connected (green), Busy (blue), Error (red).

Control	What it does
Connect / Disconnect	Open or close the connection.
Test Communication	Sends an `*IDN?`-style query and reports OK/FAILED. Enabled only when connected.
IDN line	The instrument's identification string, shown after connecting.
Clear Protection Trip (power supply only)	Clears a latched over-voltage/over-current trip. Use it if the output stays off with no other explanation.

For real hardware, pick the matching **Command profile**:

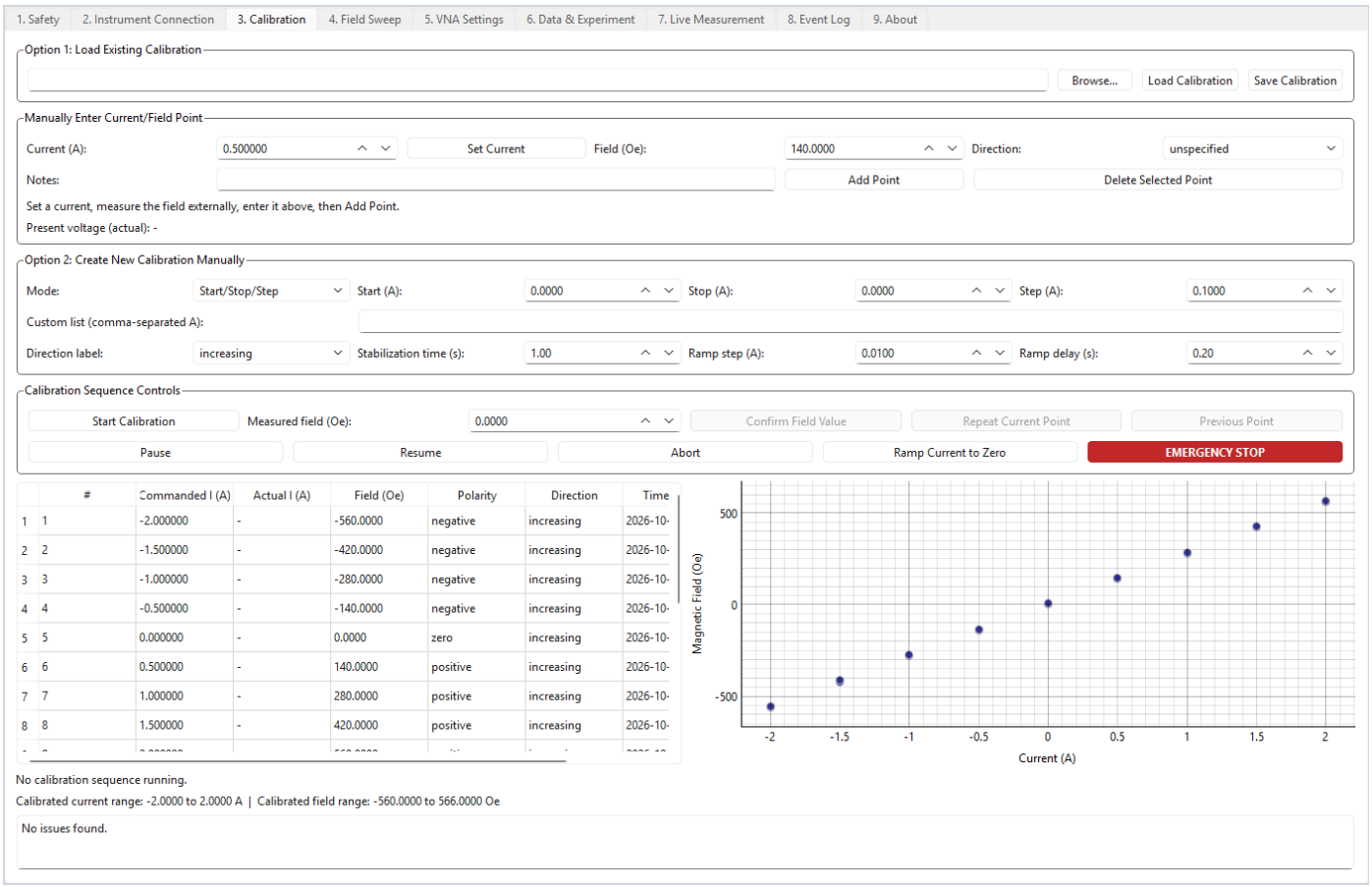
- Power supply: `AGILENT_E3634A`
- VNA: `KEYSIGHT_PNA_X_N5242B`

The default entry, `GENERIC_PLACEHOLDER`, is only a template for the simulator and for people adding a new instrument -- it will not control real hardware. (See `docs/SCPI_REPLACEMENT_GUIDE.md`.)

Once connected, the address, timeout and profile fields lock, because changing them would not affect the already-open connection. **Disconnect and reconnect** to apply a change.

On connecting the power supply, the app also sends your maximum current as the instrument's hardware current limit and re-applies the compliance voltage from tab 1 (see above).

Tab 3 -- Calibration



Purpose: teach the program how much field a given current produces. The Field Sweep tab uses this table to turn the fields you ask for into currents.

The calibration is a table of (current, field) points. A good one:

- **covers the whole field range you plan to sweep** (a sweep point outside it is rejected unless you allow extrapolation);
- includes both an **increasing** and a **decreasing** current pass if your magnet shows hysteresis, so the sweep can use the right branch;
- has at least 2 points (4 or more if you want cubic interpolation).

You can fill the table in three ways.

Option 1: Load Existing Calibration

1. Click **Browse...** and choose a `.csv` or `.json` file.
2. Click **Load Calibration**.

The file needs a current column and a field column. Accepted column names (case-insensitive): current -- `current_a`, `current`, `commanded_current_a`, `i_a`; field -- `field_oe`, `field`, `h_oe`, `magnetic_field_oe`. Optional columns: `actual_current_a` (or `actual_current`), `direction` (or `sweep_direction`), and `notes`. A current that appears twice for the same direction is reported as a duplicate and only the first is kept.

The table, the plot and the *issues* box fill in at once. Invalid points (duplicates, missing values, a non-monotonic curve, currents above the limit) are listed in the issues box and left out.

Save Calibration writes the current table to a CSV you choose. Do this after building a calibration, so you can reload it next time.

Manually Enter Current/Field Point

For when you measure the field yourself with a gaussmeter or probe.

1. Type a **Current (A)** and click **Set Current**. The supply ramps to it (same safety checks as everything else) and the supply's real output voltage is shown in *Present voltage (actual)*.
2. Measure the field with your external instrument and type it into **Field (Oe)**.
3. Pick the **Direction** (`increasing`, `decreasing` or `unspecified`), optionally add **Notes**, and click **Add Point**.
4. Repeat. Select a table row and click **Delete Selected Point** to remove a bad one.

If the instrument's measured current is more than about 10% away from what you commanded, a **Current Not Holding** warning appears -- see Troubleshooting.

> **The current stays applied after Add Point.** It is not ramped back to zero automatically. Click **Ramp Current to Zero** when you are done.

Option 2: Create New Calibration Manually (guided sequence)

The program steps through a list of currents and asks you for the field at each one.

1. Choose **Mode**:

- **Start/Stop/Step** -- fill **Start (A)**, **Stop (A)**, **Step (A)**.
- **Custom List** -- type comma-separated currents in **Custom list**, e.g. `-1, -0.5, 0, 0.5, 1`.

2. Set **Direction label** (`increasing` / `decreasing` / `unspecified`) -- it is stored with every point.
3. Set **Stabilization time (s)** (settling wait after reaching each current), **Ramp step (A)** and **Ramp delay (s)** (how gently the current is stepped; see the note below).
4. Click **Start Calibration**. Every current is checked against the maximum first; if any exceeds it, nothing starts.
5. At each point the program ramps to the current, waits, and shows: "*Enter the measured magnetic field corresponding to the presently applied current.*" Type the value in **Measured field (Oe)** and click **Confirm Field Value**. Only then does it advance.

Button	Effect
Confirm Field Value	Records the field for this current and moves on.
Repeat Current Point	Re-applies the same current without saving a point (e.g. you mis-read).
Previous Point	Goes back one point.
Pause / Resume	Hold and continue the sequence.
Abort	Stop the sequence.
Ramp Current to Zero	Ramps the supply to 0 A using the ramp step/delay above.
EMERGENCY STOP	See section 6.

When the sequence ends, the program ramps back to 0 A and disables the output.

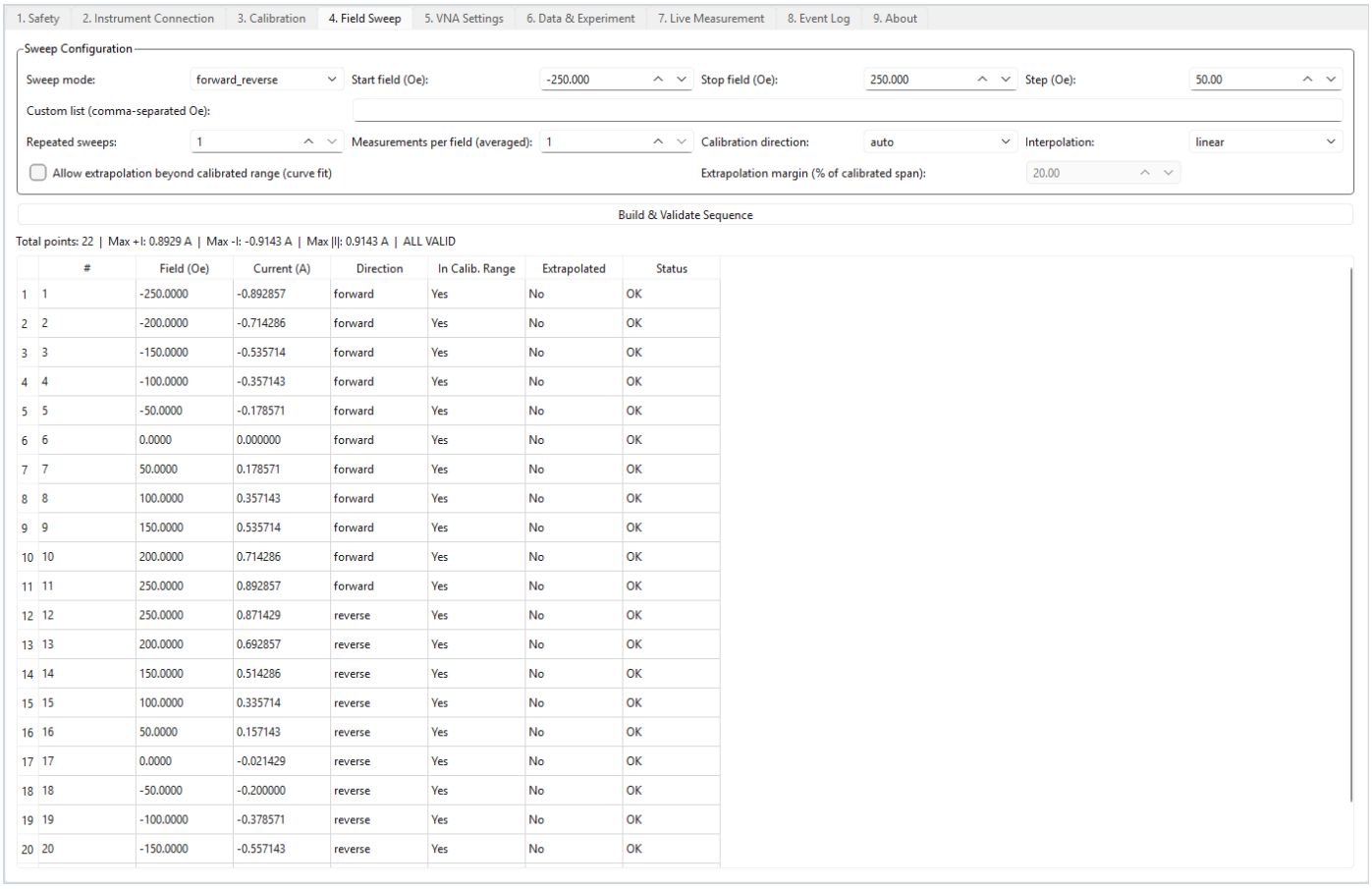
> **Why ramp step / ramp delay matter.** The coil is an inductor. A sudden jump in current induces a voltage spike that can trip the supply's over-voltage protection and drop the output to zero. Small steps with a short delay avoid this. The defaults (0.01 A step, 0.2 s delay) come from `config/app_settings.json`.

Reading the lower half of the tab

- **Table** -- every point: commanded current, actual (measured) current, field, polarity, direction, timestamp.
- **Plot** -- field vs. current for all points.
- **Calibrated current/field range** -- the span the sweep can use without extrapolating.
- **Issues box** -- `No issues found.` or a list of problems.

> Save your calibration with **Save Calibration**. When you later click **Create Experiment Folder** the current table is also copied into that experiment's `calibration/` folder.

Tab 4 -- Field Sweep



	#	Field (Oe)	Current (A)	Direction	In Calib. Range	Extrapolated	Status
1	1	-250.0000	-0.892857	forward	Yes	No	OK
2	2	-200.0000	-0.714286	forward	Yes	No	OK
3	3	-150.0000	-0.535714	forward	Yes	No	OK
4	4	-100.0000	-0.357143	forward	Yes	No	OK
5	5	-50.0000	-0.178571	forward	Yes	No	OK
6	6	0.0000	0.000000	forward	Yes	No	OK
7	7	50.0000	0.178571	forward	Yes	No	OK
8	8	100.0000	0.357143	forward	Yes	No	OK
9	9	150.0000	0.535714	forward	Yes	No	OK
10	10	200.0000	0.714286	forward	Yes	No	OK
11	11	250.0000	0.892857	forward	Yes	No	OK
12	12	250.0000	0.871429	reverse	Yes	No	OK
13	13	200.0000	0.692857	reverse	Yes	No	OK
14	14	150.0000	0.514286	reverse	Yes	No	OK
15	15	100.0000	0.335714	reverse	Yes	No	OK
16	16	50.0000	0.157143	reverse	Yes	No	OK
17	17	0.0000	-0.021429	reverse	Yes	No	OK
18	18	-50.0000	-0.200000	reverse	Yes	No	OK
19	19	-100.0000	-0.378571	reverse	Yes	No	OK
20	20	-150.0000	-0.557143	reverse	Yes	No	OK

Purpose: decide which fields to measure and prove every one of them is reachable and safe *before* anything is energised.

Sweep Configuration

Control	Meaning
Sweep mode	`forward`: start -> stop. `reverse`: stop -> start. `forward_reverse`: up, then back down (the turn-around field is measured twice -- once in each direction). `custom`: exactly the list you type.
Start field / Stop field / Step (Oe)	The range and spacing. Ignored in `custom` mode. The stop value is always included.
Custom list	Comma-separated fields in Oe, used only in `custom` mode.
Repeated sweeps	Repeats the whole list this many times.
Calibration direction	Which calibration branch converts field to current. `auto`: forward sweep points use the *increasing* branch, reverse points the *decreasing* branch, custom points use all points. `increasing` / `decreasing`: force one branch. `average`: the mean of the two branches. `all`: use every point.
Interpolation	`linear`, or `cubic` (needs 4+ calibration points on the branch being used, otherwise linear is used).

Control	Meaning
Allow extrapolation beyond calibrated range (curve fit)	Off by default. When on, a field outside your calibrated range is converted with a quadratic curve fit instead of being rejected.
Extrapolation margin (%)	How far beyond the calibrated span extrapolation is allowed (default 20%). Further out is still rejected.

> **Extrapolation warning.** Electromagnets saturate -- the field flattens at high current -- so a fitted point is less trustworthy the further it is from your measured data. Enabling it asks you to confirm. Extrapolated currents are still held to the maximum-current limit.

Build & Validate Sequence

Click it. The program converts every field to a current and fills the table:

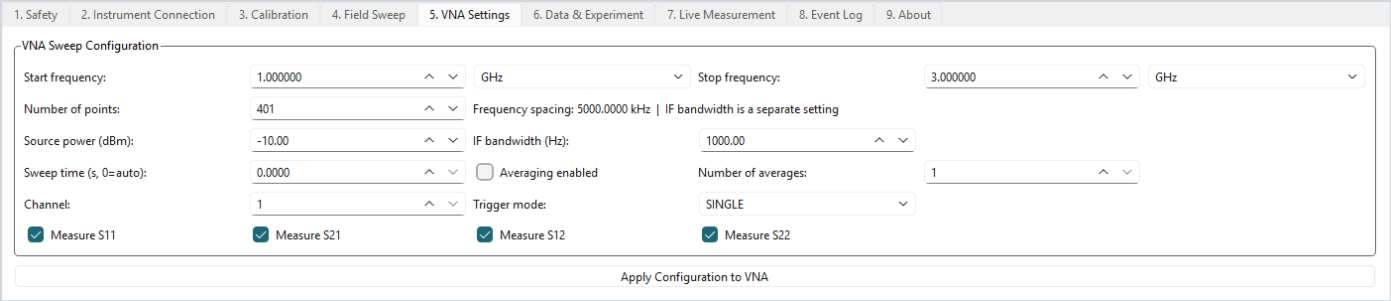
Column	Meaning
#	Position in the sequence
Field (Oe)	Requested field
Current (A)	Calculated current (`-` if it could not be computed)
Direction	`forward`, `reverse` or `custom`
In Calib. Range	Whether the field is inside the calibrated range
Extrapolated	`Yes` if it came from the curve fit (row shaded **yellow**)
Status	`OK`, or why the point is rejected (row shaded **red**)

The summary line gives the **total points**, the largest positive and negative current, the largest |I|, and either **ALL VALID** or **INVALID -- fix before starting**.

Live Measurement will not start unless the sequence is ALL VALID. To fix a red row: change the field range, calibrate further, raise the limit on tab 1 (if it is safe to do so), or enable extrapolation -- then click **Build & Validate Sequence** again.

> **The built sequence is a snapshot.** If you change the calibration or any sweep setting afterwards, click **Build & Validate Sequence** again.

Tab 5 -- VNA Settings



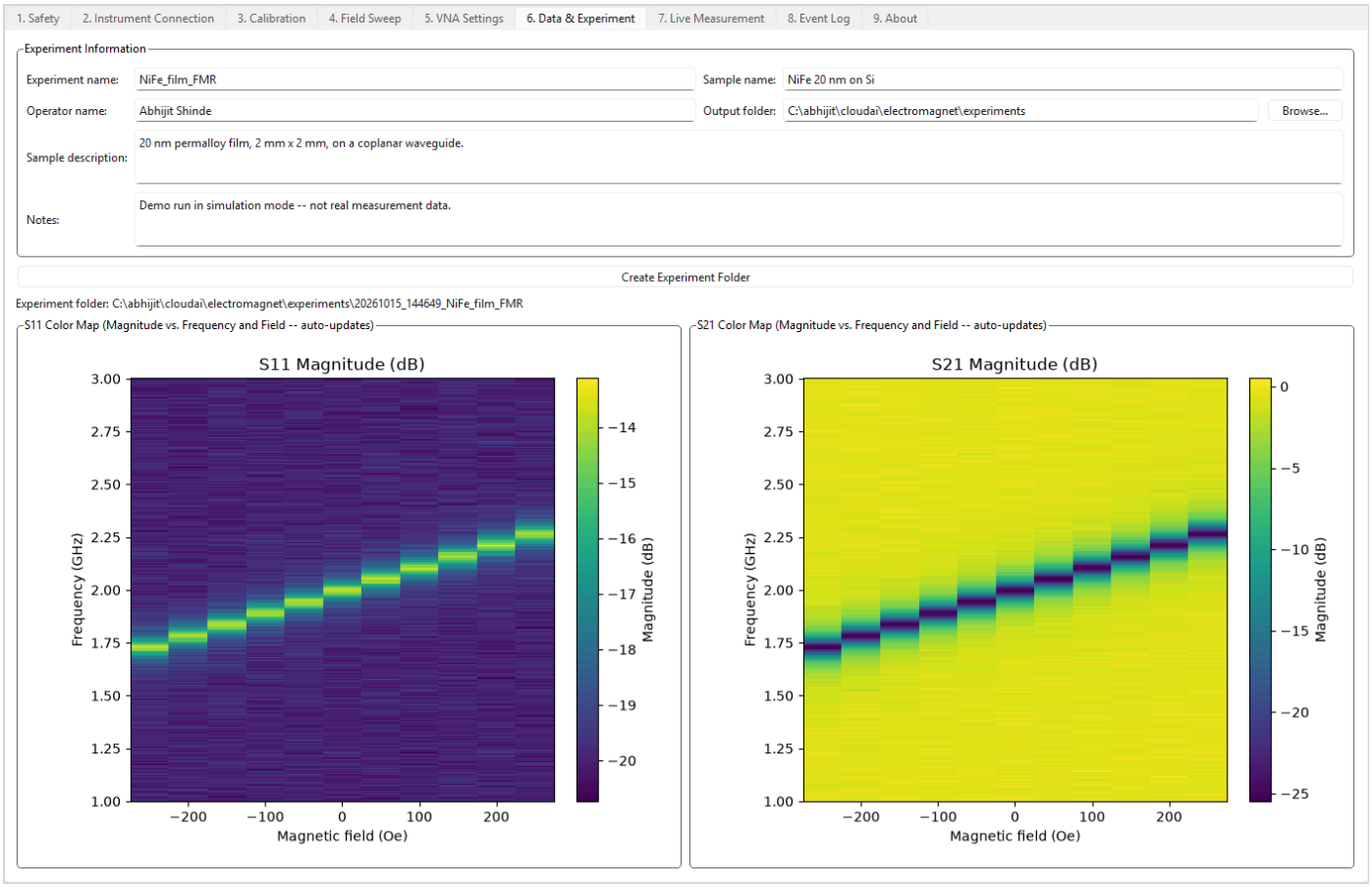
Purpose: configure the VNA sweep.

Control	Meaning
Start frequency / Stop frequency	Value plus a unit drop-down (Hz, kHz, MHz, GHz, THz). Changing the unit converts the number so the frequency does not change.
Number of points	2 - 100001. The frequency spacing (kHz) is shown next to it.
Source power (dBm)	VNA output power, -60 to +30 dBm. Keep it low enough not to drive your sample nonlinear.
IF bandwidth (Hz)	Receiver bandwidth. Lower = less noise but a slower sweep.
Sweep time (s, 0=auto)	Fixed sweep time, or `0` to let the VNA choose.
Averaging enabled / Number of averages	VNA-side averaging for lower noise (slower).
Channel	VNA channel number (1-32).
Trigger mode	`SINGLE`, `CONTINUOUS` or `EXTERNAL`. Leave on `SINGLE` for normal use -- the app triggers one sweep per field point.
Measure S11 / S21 / S12 / S22	Which S-parameters to acquire (all on by default). Unchecked ones are not read, not plotted and not saved.
Apply Configuration to VNA	Stores the settings and, if the VNA is connected, pushes them to the instrument.

> **Apply after connecting.** The settings only reach the instrument when you click **Apply Configuration to VNA** *while the VNA is connected*. If you click it before connecting you only get "Configuration saved. Connect the VNA to apply it." -- connect, then click it again. Click it again after any change (including toggling the S-parameter checkboxes) and after every reconnect. Without it, a measurement fails with "VNA has not been configured".

The app does **not** perform a VNA calibration. Do any calibration (e.g. a full 2-port calibration, which you want for meaningful S12/S22) on the instrument itself before the experiment.

Tab 6 -- Data & Experiment



Purpose: name the experiment, create its folder, and see the field-vs- frequency color maps build up.

Experiment Information

Field	Meaning
Experiment name	Used in the folder name.
Sample name	Free text, saved to `metadata.json`.
Operator name	Free text, saved to `metadata.json`.
Output folder	Where experiment folders are created. Default is `experiments/` next to the program. **Browse...** to change.
Sample description / Notes	Free text, saved to `metadata.json`.

Click **Create Experiment Folder**. This makes a new timestamped folder (`YYYYMMDD_HHMMSS_<experiment name>`) with its sub-folders, writes `metadata.json`, records the maximum current, and copies the current calibration table into it. The full path is shown under the button.

Do this before you start a measurement or a background measurement -- they refuse to run without an experiment folder. Clicking the button again starts a *new* folder (the old one is left untouched on disk) and resets the color maps.

Color maps

Two square plots, **S11** and **S21**, show magnitude (dB) with *field on the x-axis and frequency on the y-axis*. They **update by themselves** after each field point is saved -- there is no button to press. Before any data exists they read "no data yet".

Color maps for **all four** S-parameters (S11, S21, S12, S22) are saved to the experiment's `plots/` folder as they update, even though only S11 and S21 are shown on screen.

Tab 7 -- Live Measurement



Purpose: run the experiment and watch it.

Live Status panel

Item	Shows
Power Supply / VNA	Connection state
Max current	The limit from tab 1
Present current	Target current, and the measured one once a point completes
Present voltage (actual)	The supply's *real metered* output voltage (not the compliance setpoint, which a front panel may display instead)

Item	Shows
Present field	Requested field of the current point
Direction	`forward`, `reverse` ...
VNA sweep status	The run state: `Running`, `Paused`, `Completed`, `Stopped` or `Error`
Output folder	The experiment folder (or a reminder to create one)
Elapsed	Time since Start Measurement
Progress bar	Points done / total
Red warning line	The last error message, if any

The four plots

Top-left **S11**, top-right **S21**, bottom-left **S12**, bottom-right **S22** -- all magnitude in dB versus frequency in GHz. Each point you measure adds a trace, labelled `H=<field> Oe` in the legend beside the plot (10 rows are visible; scroll the legend for more).

- Mouse: drag to pan, scroll wheel to zoom, right-click for the plot menu.
- **Overlay traces** -- checked: keep every trace so you can compare fields. Unchecked (default): show only the latest trace. Toggling it clears the plots.
- **Clear Plots** -- remove all traces from the screen (saved data is not affected).
- **Autoscale** -- reset the axes to fit.

Phase is **not** plotted, but it *is* saved for every S-parameter in the CSV and HDF5 files.

Ramp & Stabilization

Control	Meaning
Ramp step (A)	Size of each current step when moving between points.
Ramp delay (s)	Pause between steps.
Stabilization (s)	Extra wait after reaching each point's current, before the VNA sweep.

Defaults come from `config/app_settings.json`. The coil's inductance means smaller steps / longer delays are gentler; larger ones are faster but risk tripping the supply (see tab 3).

Measure Background (0A)

Saves a reference measurement with **no applied field**, separate from the sweep:

1. The program asks you to confirm.
2. It ramps the current to 0 A, enables the output, triggers one VNA sweep, and reads the enabled S-parameters.
3. It saves them to `background/background.csv` and as a `background` group in the HDF5 file.

It does **not** disturb the field-sweep data, color maps or cumulative graphs. It needs both instruments connected, an experiment folder, and a configured VNA, and cannot be used while a sweep is running. The output is left

enabled at 0 A afterwards.

Typical use: measure it once, before the sweep, so you can later subtract or normalise the field data against it.

Start Measurement

Click **Start Measurement**. The program first checks everything and, if any condition fails, lists *all* the problems in one dialog:

- maximum current is set;
- power supply and VNA both connected;
- calibration present and valid;
- a field sweep has been built and is **ALL VALID**;
- an experiment folder has been created;
- the VNA has been configured;
- a compliance voltage has been sent to the power supply (needed whenever the sweep uses any non-zero current).

It then asks "N field point(s) will be measured ... Proceed?". After you confirm, for each point in turn the program:

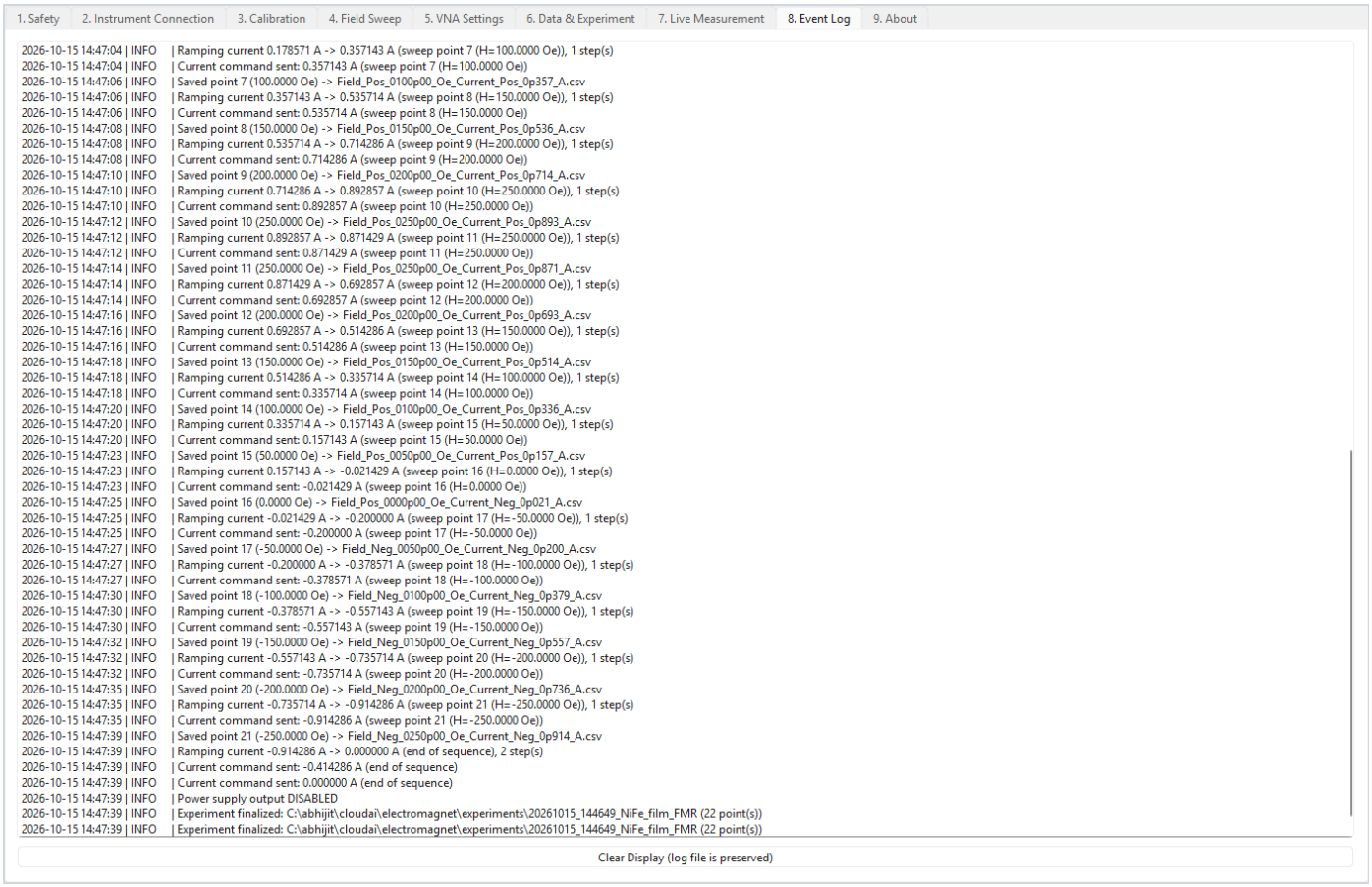
1. ramps the current to the point's value (in the ramp step/delay you set),
2. waits the stabilization time,
3. triggers one VNA sweep and reads S11/S21/S12/S22,
4. reads the actual current and voltage,
5. **writes the point to disk immediately** (per-point CSV, combined CSV, HDF5, cumulative graphs, color maps), and
6. adds the traces to the plots.

At the end it ramps to 0 A, disables the output, and shows "Measurement Finished" with the number of points completed.

Run controls

Button	Effect
Pause / **Resume**	Hold the sequence at the current point and continue later. The current stays applied while paused.
Abort	Stop the sequence now. The program ramps to zero, disables the output, and keeps all data measured so far.
Ramp to Zero	Ramp the supply to 0 A using the ramp step/delay above (e.g. after a pause).
EMERGENCY STOP	See section 6.

Tab 8 -- Event Log



A read-only, live list of everything the program does: connections, every current command, saved points, pauses/aborts/emergency stops, warnings and errors, file saves, and shutdown.

Clear Display only empties this view. The complete log is always kept in `logs/application.log` and, once you create an experiment, in that experiment's `event_log.log`.

When something unexpected happens, this is the first place to look.

Tab 9 -- About



Credits and a one-paragraph description of the tool. Always accessible, even before the maximum current is set.

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6. Stopping, aborting and emergencies

Situation	Use	What happens
You want to stop for a moment	**Pause**	Sequence holds; current stays applied. **Resume** continues.
You want to end the run but keep the data	**Abort**	The program ramps to zero, disables the output, and keeps all points saved so far.
You want the current off but stay in the program	**Ramp to Zero** / **Ramp Current to Zero**	Smooth ramp to 0 A.
Something is wrong *right now*	**EMERGENCY STOP** (red button, Calibration and Live tabs)	Blocks **all** further current commands, stops the running process, and attempts to disable the output. Confirm on the front panel and in the Event Log that the output is off.
Output voltage exceeds the monitoring threshold	(automatic)	The app triggers the same emergency stop on its own and tells you with a **VOLTAGE EMERGENCY STOP** dialog.

After an emergency stop, restart the application. The Safety tab shows `EMERGENCY STOP ACTIVE`, and the program rejects every current command until it is restarted -- there is no on-screen button that clears it. Check the Event Log and the instrument's front panel to confirm the output really is off before you do anything else.

Closing the window while a process is running or the output is enabled asks for confirmation, then stops the process, ramps to zero, disables the output, saves all data and disconnects both instruments. It never just quits

with the magnet energised. If it could not *confirm* the shutdown (for example GPIB communication was lost) it says so in the log -- then check the instrument manually.

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7. Where your data goes

Each experiment gets its own folder under the output folder:

```
experiments/20261015_144649_NiFe_film_FMR/
  metadata.json                  experiment/sample/operator info, max current,
                                 start and finish time, number of points
  event_log.log                  this experiment's log
  calibration/calibration.csv    copy of the calibration used
  raw/Field_Pos_0100p00_Oe_Current_Pos_0p357_A.csv
                                 one CSV per field point (name encodes the
                                 field and current), written immediately
  processed/combined_data.csv    every point so far, in one table
  processed/experiment_data.h5   HDF5: one group per point + "background"
  background/background.csv      the 0 A reference (if measured)
  graphs/s11_magnitude.png       cumulative overlay of all points, per
  graphs/s21_magnitude.png       S-parameter (magnitude only), refreshed
  graphs/s12_magnitude.png       after every point
  graphs/s22_magnitude.png
  plots/s11_colormap.png         field-vs-frequency color maps, all four
  plots/s21_colormap.png         S-parameters, refreshed after every point
  plots/s12_colormap.png
  plots/s22_colormap.png
```

CSV columns (per-point, combined and background files). One row per frequency point; a block of five columns for each S-parameter you enabled:

Column	Meaning
`frequency_hz`	Frequency of the row
`requested_field_oe`	Field you asked for (not in the background file)
`current_a`	Commanded current
`actual_current_a`, `actual_voltage_v`	Measured current and voltage (blank if unavailable)
`direction`, `sweep_number`	`forward` / `reverse` / `custom`; repeat index (not in background)
`timestamp`	When the point was taken
`s11_real`, `s11_imag`	Complex S11
`s11_magnitude`	Linear magnitude

Column	Meaning
`s11_magnitude_db`	Magnitude in dB
`s11_phase_deg`	Phase in degrees
`s21_*`, `s12_*`, `s22_*`	Same five columns for S21, S12, S22

HDF5 layout: each field point is a group `point_0000`, `point_0001`, ... with the field, current, direction, etc. as attributes and datasets `frequency_hz` and `<sparam>_real` / `<sparam>_imag`. The 0 A reference is a group named `background`.

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8. Troubleshooting

Symptom / message	Likely cause and fix
Tabs 2-7 are greyed out	The maximum current has not been set. Go to tab 1 and click **Set / Confirm Maximum Current**.
Cannot Change Maximum Current / the limit field is greyed	A process is running or the output is enabled. Abort, **Ramp Current to Zero**, disable the output.
Connection Failed	Wrong GPIB address, instrument off, cable/adapter problem, or the wrong **Command profile**. Check `docs/GPIB_ADDRESS_GUIDE.md`; try **Simulation Mode** to confirm the program itself works.
Connects, but nothing responds correctly on real hardware	The **Command profile** is still `GENERIC_PLACEHOLDER`. Choose `AGILENT_E3634A` / `KEYSIGHT_PNA_X_N5242B`, disconnect and reconnect.
Invalid Voltage Value (0 V)	Coil resistance is 0 or no maximum current is set. Fix, click **Calculate from Max Current**, then set again.
Current Not Holding	The commanded current is not appearing at the instrument. Usually a protection trip (OVP/OCP) or too-low compliance voltage. Click **Clear Protection Trip** on tab 2, and re-click **Set Monitoring Threshold & Compliance Voltage** on tab 1 with the supply connected. Use smaller ramp steps.
Output drops to ~0 V right after a large current step	The inductive spike tripped the supply's OVP. Reduce **Ramp step** / increase **Ramp delay**, then **Clear Protection Trip**.
Cannot Start Measurement	The dialog lists exactly what is missing (limit, connections, calibration, sweep, experiment folder, VNA configuration, compliance voltage). Fix each item.
Sweep table has red rows	A field is outside the calibrated range or needs more than the maximum current. See Tab 4.
VNA has not been configured	Click **Apply Configuration to VNA** on tab 5 *while the VNA is connected* (see the note there).
Timed out waiting for VNA sweep to complete	The sweep takes longer than 30 s. Use fewer points, a higher IF bandwidth, or less averaging.

Symptom / message	Likely cause and fix
No experiment folder created	Tab 6 -> **Create Experiment Folder**.
Color maps say "no data yet"	No field point has been saved in this experiment folder yet.
Plots look empty after switching **Overlay**	Toggling overlay clears the plots; new points will appear as they are measured.
EMERGENCY STOP ACTIVE on tab 1	Restart the application (see section 6).
Something odd happened	Read tab 8 (Event Log) or `logs/application.log`.

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9. Known limitations

- **"Measurements per field (averaged)" on the Field Sweep tab is not yet used** by the measurement code -- changing it has no effect. Use the VNA's own **Averaging** on tab 5 instead.
- **The Agilent E3634A is a single-quadrant (unipolar) supply** -- it cannot source negative current by itself. A sweep through negative fields needs external polarity reversal (a relay or manual reversal of the coil leads). See `docs/SCPI_REPLACEMENT_GUIDE.md`.
- The calibration table is **not auto-saved** while you build it. Click **Save Calibration** (or create an experiment folder, which copies it).
- A measurement runs one VNA sweep per field point; the sweep must finish within 30 s.
- The maximum current and the voltage threshold are intentionally **not** remembered between sessions; coil resistance and margin are.

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10. Glossary

Term	Meaning
Oe (oersted)	Unit of magnetic field strength used throughout the app (1 Oe ≈ 79.58 A/m).
A	Ampere -- the coil current.
S11, S22	Reflection coefficients at VNA port 1 / port 2.
S21, S12	Transmission coefficients port 1 -> 2 / port 2 -> 1.
dB	Decibels; magnitude plots show 20·log10 of the linear magnitude of S.
IFBW	Intermediate-frequency bandwidth of the VNA receiver.

Term	Meaning
Compliance voltage	The voltage ceiling of a constant-current supply; the supply raises its voltage up to this limit to push the commanded current through the coil.
OVP / OCP	The instrument's over-voltage / over-current protection circuits.
Background	A measurement taken at 0 A (no applied field), used as a reference.
Hysteresis	The field produced at a given current depends on whether the current is increasing or decreasing; hence separate increasing/decreasing calibration branches.
Simulation Mode	Software-simulated instruments; no hardware is driven.